

LAB EXPERIMENT ITA0506-COMPUTER VISION

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EXPNO: 31. Morphological Operation Using Opening

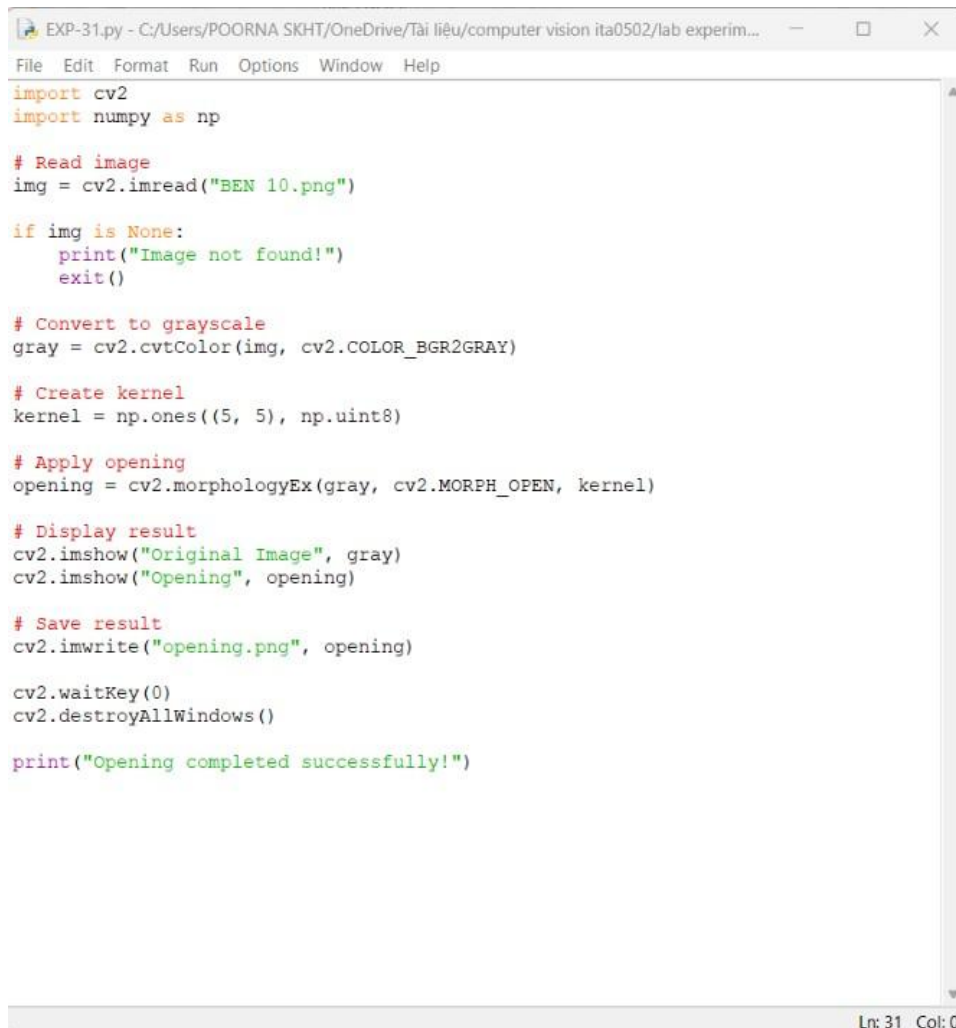
Aim

To perform morphological opening on an image using OpenCV.

Algorithm

1. Import OpenCV.
2. Read BEN 10.png.
3. Convert the image to grayscale.
4. Create a kernel.
5. Apply opening using cv2.morphologyEx().
6. Display and save the result.

CODE:

A screenshot of a Python IDE window titled 'EXP-31.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...'. The window contains a menu bar with 'File', 'Edit', 'Format', 'Run', 'Options', 'Window', and 'Help'. The code is as follows:

```
import cv2
import numpy as np

# Read image
img = cv2.imread("BEN 10.png")

if img is None:
    print("Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply opening
opening = cv2.morphologyEx(gray, cv2.MORPH_OPEN, kernel)

# Display result
cv2.imshow("Original Image", gray)
cv2.imshow("Opening", opening)

# Save result
cv2.imwrite("opening.png", opening)

cv2.waitKey(0)
cv2.destroyAllWindows()

print("Opening completed successfully!")
```

The status bar at the bottom right shows 'Ln: 31 Col: 0'.

OUTPUT :

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-26.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-28.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-29.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-30.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-31.py
```



Result

Morphological opening is successfully performed. It helps remove small unwanted noise from the image.